

Artificial Intelligence

Artificial Intelligence AI is the simulation of human intelligence in machines, enables them to learn, reason, solve problems, and make decisions.

Different from traditional programming — Unlike traditional programming, where developers write explicit rules, AI systems learn pattern from data to make predictions or decisions.

Types

Narrow AI

General AI

Super AI

Designed to perform a specific task, such as voice assistants or recommendations system.

A theoretical AI, capable of performing any intellectual task a human can do.

A hypothetical AI that passes beyond human intelligence in every aspect.

examples: ChatGPT, Face recognition, Recommendation systems, self-driving systems.

Machine learning

Machine learning is the subset of Artificial Intelligence that enables computers to learn patterns from data & Make predictions or decisions without being explicitly programmed.

How works:

- Collect data
- Train a Model
- Evaluate performance
- Make predictions on new data
- Improve with more data

Types

Semi-Supervised Learning

uses small amount of labeled data with large amount of unlabeled data.

Supervised Learning

Leads from Labeled data

Unsupervised learning

learns from unlabeled data by finding patterns

Reinforcement learning

Leads through rewards and penalties

Detailed :-

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Supervised Learning
Supervised Learning trains a model using labeled data, where both input and expected output are known.

Common applications:

- Spam email detection
- Customer churn prediction
- Image classification
- House price prediction

Popular algorithms include:

- Linear regression
- Logistic regression
- Decision Tree
- Random forest
- Support Vector Machine (SVM)

Unsupervised Learning
Unsupervised Learning works with unlabeled data to discover hidden patterns or group similar data points.

Applications:

- Customer Segmentation
- Market Basket Analysis
- Anomaly detection
- Recommendation systems

Popular Algorithm :

- K Means
- DBSCAN
- Hierarchical Clustering
- PCA

Reinforcement learning

Reinforcement Learning RL is a Learning method where an agent interacts with an environment and learns by receiving rewards or penalties.

Unlike SL, RL does not require labelled data. Instead, It learns through trial and error to maximize cumulative rewards.

Applications :

- Robotics
- Self-driving cars
- Game AI
- Resource optimization

Semi supervised Learning

Semi supervised Learning is a hybrid branch of machine learning sits between supervised learning (which requires entirely labeled data) and unsupervised learning.

Applications :

- Medical Diagnosis
- Speech and Language Analysis
- Web content classification

Deep Learning

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple hidden layers to learn complex patterns from large datasets.

Applications:

- Image recognition, speech recognition, NLP, autonomous vehicles.

Generative AI

Generative AI is a type of artificial intelligence technology that can produce various types of content, including text, imagery, audio and synthetic data.

example

- | | | |
|-------|---|------------------|
| Text | → | ChatGPT |
| Image | → | Image generators |
| Audio | → | Audio generators |
| Code | → | Code generators |
| Video | → | Video generators |

Difference of AI Engineer

Core focus - Building user facing, AI-powered applications by integrating pre-trained foundational models (like LLMs).

What they do daily - Connect applications to model endpoints, build Retrieval Augmented Generation (RAG) pipelines, set up vector databases, and design autonomous AI-agents.

Key skills - Software engineering, API integration, prompt engineering, vector databases, and agent frameworks (LangChain).

AI Researcher

Core focus - Inventing brand-new algorithms, neural network architectures, or theoretical frameworks to push the boundaries of what AI can do.

What they do daily - Read academic papers, design novel machine learning models from scratch, run heavy experimental training, publish findings at conferences.

Key skills - Advanced mathematics, deep learning, PyTorch / TensorFlow, and high performance computing.

ML Engineer

Core focus - Taking experimental machine learning models and making them run reliably in productions.

What they do daily - Package models to APIs, write automated retraining scripts, optimize model performance, and set up monitoring tools to track data drift and scaling.

key skills - Advanced Python, Docker, Kubernetes, CI/CD tools, MLOps frameworks (MLflow), cloud services.

Data Engineer

Core focus - Building and managing the underlying infrastructure that moves and stores data.

What they do daily - Design scalable data pipelines, manage data warehouse / lakehouses, clean incoming data streams, and ensure data flows safely from sources to destinations.

key skills - SQL, Apache Spark, Kafka, Airflow, cloud platforms (AWS, Azure, GCP), and database architecture.

Data Analyst

Core focus - Translating historical data into clear business insights, charts, and reports.

What they do daily - Write SQL queries, clean spreadsheets, build interactive dashboards (Tableau / Power BI), and track key performance Indicators (KPIs) for communication management.

Key skills - Excel, SQL, basic Python, data visualisation, and business communication.

Git

Git is fast
Git is built to work on the Linux kernel, meaning that it was built to handle repositories with tens of millions of lines of code from the start. Speed and performance has always been a primary design goal of Git.

Git is widely used
According to the 2022 Stack Overflow developer use Git.

Git has a huge ecosystem of tools
The core Git project is just a command-line tool, but Git exploded in popularity

in the early 2010s thanks to Git hosting services like GitLab, GitHub, and more.

Since Git was created, many GUIs, editors, integrations, and command line tools have been built to make working with Git more convenient. ~~Your~~ fa

Difference between Git and GitHub —

	Git	GitHub
What it is	A local software command-line software tool.	A web-based cloud platform.
Primary Purpose	Tracks code history and manages versions.	Hosts Git repositories and enables teamwork.
Where it runs	Locally understand on your personal computer	Online in the cloud.
Internet Required?	No works completely offline	Yes requires an internet connection
Interface	Command Line Interface (CLI) by default.	Graphical User Interface (GUI) website.
Maintained By	The open-source Linux community.	Microsoft
Competitors	Mercurial, SVN, Perforce	GitLab, Bitbucket